WORKED SOLUTION

# STEP III 1989, Question 7

Pure • Shear Matrices

**STEP III 1989, Question 7**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q7

## The key idea

Everything is easiest in terms of two unit vectors: $\mathbf{u}$ along the line $L\colon y = x\tan\alpha$, and $\mathbf{n}$ perpendicular to it. The shear moves each point parallel to $\mathbf{u}$ by an amount proportional to its signed distance from $L$, so $\mathbf{M} = \mathbf{I} + k\,\mathbf{u}\mathbf{n}^{\mathsf{T}}$. For the second part, a matrix commutes with $\mathbf{I} + k\mathbf{A}$ exactly when it commutes with $\mathbf{A}$, which strips out the clutter.

## Solution

Write $s = \sin\alpha$ and $c = \cos\alpha$. Since $\alpha$ is acute, $s > 0$ and $c > 0$. Let

$$\mathbf{u} = \begin{pmatrix} c \\ s \end{pmatrix}, \qquad \mathbf{n} = \begin{pmatrix} -s \\ c \end{pmatrix},$$

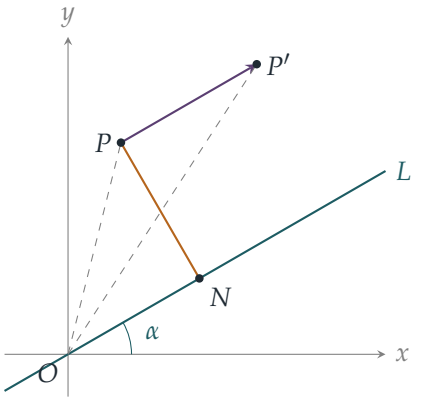
so $\mathbf{u}$ is the unit vector along $L$ at angle $\alpha$ to the $x$-axis, and $\mathbf{n}$ is the unit normal pointing to the upper-left side of $L$.

### Finding $\mathbf{M}$

For $P = (x, y)$ let

$$d = \mathbf{n} \cdot \overrightarrow{OP} = y\cos\alpha - x\sin\alpha,$$

the *signed* distance from $P$ to $L$. Then $PN = |d|$, and $d > 0$ exactly when $P$ lies above $L$.




Take $P$ above $L$, so $d > 0$. By (i), $\overrightarrow{PP'}$ points along $\mathbf{u}$, so $\overrightarrow{PP'} = \lambda\mathbf{u}$ with $\lambda > 0$. Condition (ii) is consistent with this: the two-dimensional cross product is

$$\overrightarrow{OP} \times \overrightarrow{OP'} = \overrightarrow{OP} \times (\overrightarrow{OP} + \lambda\mathbf{u}) = \lambda\,(xs - yc) = -\lambda d < 0,$$

so the turn from $OP$ to $OP'$ is clockwise, as required. Condition (iii) gives $\lambda = k\,PN = kd$. (For these conditions to hold together we need $k > 0$.) Hence, for every $P$ above $L$,

$$\overrightarrow{OP'} = \overrightarrow{OP} + kd\,\mathbf{u} = \overrightarrow{OP} + k\,\mathbf{u}\,(\mathbf{n} \cdot \overrightarrow{OP}).$$

**Ask yourself** We have only used points above $L$. Why does that fix $\mathbf{M}$ completely?

T is linear, and so is the map $\mathbf{v} \mapsto \mathbf{v} + k\,\mathbf{u}(\mathbf{n} \cdot \mathbf{v})$. They agree on the open half-plane above $L$, which contains two linearly independent vectors (for example $\mathbf{n}$ and $\mathbf{n} + \mathbf{u}$). Two linear maps that agree on a basis are equal. So for *all* vectors,

$$\mathbf{M} = \mathbf{I} + k\,\mathbf{u}\mathbf{n}^{\mathsf{T}} = \mathbf{I} + k\begin{pmatrix} c \\ s \end{pmatrix}\begin{pmatrix} -s & c \end{pmatrix} = \mathbf{I} + k\begin{pmatrix} -sc & c^2 \\ -s^2 & sc \end{pmatrix}.$$

That is,

$$\mathbf{M} = \begin{pmatrix} 1 - k\sin\alpha\cos\alpha & k\cos^2\alpha \\ -k\sin^2\alpha & 1 + k\sin\alpha\cos\alpha \end{pmatrix},$$

as required. As a sanity check, $\mathbf{n} \cdot \mathbf{u} = 0$ gives $\mathbf{M}\mathbf{u} = \mathbf{u}$, so $L$ is fixed pointwise. Also $\det\mathbf{M} = 1 - k^2s^2c^2 + k^2s^2c^2 = 1$, as a shear should have.

### The commuting condition

Let $\mathbf{A} = \mathbf{u}\mathbf{n}^{\mathsf{T}} = \begin{pmatrix} -sc & c^2 \\ -s^2 & sc \end{pmatrix}$ and $\mathbf{X} = \begin{pmatrix} p & q \\ r & t \end{pmatrix}$. Since $\mathbf{M} = \mathbf{I} + k\mathbf{A}$ and $\mathbf{I}$ commutes with everything,

$$\mathbf{X}\mathbf{M} - \mathbf{M}\mathbf{X} = k(\mathbf{X}\mathbf{A} - \mathbf{A}\mathbf{X}).$$

Because $k \neq 0$, $\mathbf{X}$ commutes with $\mathbf{M}$ if and only if it commutes with $\mathbf{A}$. Multiplying out,

$$\mathbf{XA} = \begin{pmatrix} -psc - qs^2 & pc^2 + qsc \\ -rsc - ts^2 & rc^2 + tsc \end{pmatrix}, \qquad \mathbf{AX} = \begin{pmatrix} -psc + rc^2 & -qsc + tc^2 \\ -ps^2 + rsc & -qs^2 + tsc \end{pmatrix},$$

so

$$\mathbf{XA} - \mathbf{AX} = \begin{pmatrix} -(qs^2 + rc^2) & -c^2(t-p) + 2qsc \\ -s^2(t-p) - 2rsc & qs^2 + rc^2 \end{pmatrix}.$$

This is zero if and only if

$$\begin{aligned} &\text{(a)} \quad qs^2 + rc^2 = 0, \\ &\text{(b)} \quad c^2(t-p) = 2qsc, \\ &\text{(c)} \quad s^2(t-p) = -2rsc. \end{aligned}$$

Since $c \neq 0$, (b) is $t - p = 2q\tan\alpha$. Since $s \neq 0$, (c) is $t - p = -2r\cot\alpha$.

**Ask yourself** There are three equations but the stated condition contains only two. What happened to (a)?

It follows from the other two. If $2q\tan\alpha = -2r\cot\alpha$, multiplying by $\frac{1}{2}sc$ gives $qs^2 = -rc^2$, which is (a). So

$$\mathbf{XM} = \mathbf{MX} \iff t - p = 2q\tan\alpha = -2r\cot\alpha.$$

Both directions hold: every step above is an equivalence. This is the required necessary and sufficient condition.

## Checks

- With $\alpha = \frac{\pi}{4}$, $k = 2$ we get $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$. The point $(0,1)$, at distance $\frac{1}{\sqrt{2}}$ above $y = x$, goes to $(1,2)$. That is a move of $\sqrt{2} = 2 \times \frac{1}{\sqrt{2}}$ in direction $45°$, and the turn is clockwise.
- The condition is satisfied by $\mathbf{X} = \mathbf{M}$ itself: $t - p = 2ksc$, $2q\tan\alpha = 2kc^2 \cdot \frac{s}{c} = 2ksc$ and $-2r\cot\alpha = 2ks^2 \cdot \frac{c}{s} = 2ksc$. It is also satisfied by $\mathbf{I}$, with all three expressions zero.
- Numerically, for random $\alpha$, $k$ and $P$ above $L$, $\mathbf{M}$ reproduced (i)–(iii). Random matrices satisfying the condition commuted with $\mathbf{M}$ to within $10^{-8}$, and random matrices violating it did not.

## Common slips

- Using the unsigned distance $PN$ for every point. The displacement is $k$ times the *signed* distance. Points below $L$ move the other way, which is what makes T linear.
- Proving only "if" (or only "only if") for the commuting condition, or forgetting to show that the $(1,1)$ entry gives nothing new.

## Beyond STEP

The matrices that commute with $\mathbf{M}$ here are exactly those of the form $a\mathbf{I} + b\mathbf{A}$: check that $t - p = 2q\tan\alpha = -2r\cot\alpha$ is a two-parameter family. This is an instance of a general fact. For a $2 \times 2$ matrix that is not a multiple of $\mathbf{I}$, the matrices commuting with it are exactly $a\mathbf{I} + b\mathbf{M}$. For $n \times n$ matrices, the analogue (commuting matrices are polynomials in $\mathbf{M}$) holds when the minimal polynomial equals the characteristic polynomial. Strang's *Introduction to Linear Algebra* develops rank-one matrices $\mathbf{u}\mathbf{n}^{\mathsf{T}}$ and the ideas behind this.

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